



ENGINEERING MATHEMATICS - II

Dr. G. K. Jagatheswari

Department of Science and Humanities

ENGINEERING MATHEMATICS - II

UNIT 5 : Fourier series

Session : 7

Sub Topic : Practical Harmonic Analysis

Department of Science and Humanities

ENGINEERING MATHEMATICS -II

Practical Harmonic Analysis

In many engineering and scientific problems, $f(x)$ is not given directly, rather set of discrete values of function are given in the form (x_i, y_i) , $i = 1, 2, 3, \dots, m$ where x_i 's are equispaced. The process of obtaining $f(x)$ in terms of Fourier series from given set of values (x_i, y_i) , is known as practical harmonic analysis.

x	x_1	x_2	x_3				x_N
$y = f(x)$	y_1	y_2	y_3				y_N

Recall!!!

We know that, for a periodic function $f(x)$ of period 2π , defined over $(0, 2\pi)$, the Fourier series expansion is given by

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx) \quad \dots \dots \dots (1)$$

$$a_0 = \frac{1}{\pi} \int_0^{2\pi} f(x) dx$$

$$a_n = \frac{1}{\pi} \int_0^{2\pi} f(x) \cos nx dx \quad \text{for } n = 1, 2, 3 \dots$$

$$b_n = \frac{1}{\pi} \int_0^{2\pi} f(x) \sin nx dx \quad \text{for } n = 1, 2, 3 \dots$$

Equation (1) can be rearranged as

$$f(x) = \frac{a_0}{2} + (a_1 \cos x + b_1 \sin x) + (a_2 \cos 2x + b_2 \sin 2x) + (a_3 \cos 3x + b_3 \sin 3x) + \dots$$

The term $(a_1 \cos x + b_1 \sin x)$ is called the **First Harmonic**.

The term $(a_2 \cos 2x + b_2 \sin 2x)$ is called the **Second Harmonic**.

The term $(a_3 \cos 3x + b_3 \sin 3x)$ is called the **Third Harmonic** and so on.

Fourier series of functions having arbitrary period $2l$

Recall!!!

The Fourier series expansion of a function $f(x)$ defined over an interval $(0, 2l)$ is given by

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left(a_n \cos \frac{n\pi x}{l} + b_n \sin \frac{n\pi x}{l} \right) \dots \dots \dots (2)$$

where

$$a_0 = \frac{1}{l} \int_0^{2l} f(x) dx.$$

$$a_n = \frac{1}{l} \int_0^{2l} f(x) \cos \frac{n\pi x}{l} dx$$

$$b_n = \frac{1}{l} \int_0^{2l} f(x) \sin \frac{n\pi x}{l} dx$$

Equation (2) can be rearranged as

$$f(x) = \frac{a_0}{2} + (a_1 \cos \frac{\pi x}{l} + b_1 \sin \frac{\pi x}{l}) + (a_2 \cos \frac{2\pi x}{l} + b_2 \sin \frac{2\pi x}{l}) + (a_3 \cos \frac{3\pi x}{l} + b_3 \sin \frac{3\pi x}{l}) + \dots$$

The term $(a_1 \cos \frac{\pi x}{l} + b_1 \sin \frac{\pi x}{l})$ is called the **First Harmonic**.

The term $(a_2 \cos \frac{2\pi x}{l} + b_2 \sin \frac{2\pi x}{l})$ is called the **Second Harmonic**.

The term $(a_3 \cos \frac{3\pi x}{l} + b_3 \sin \frac{3\pi x}{l})$ is called the **Third Harmonic** and so on.

ENGINEERING MATHEMATICS -II

Practical Harmonic Analysis



Suppose we have a set of N values of $y = f(x)$ having period 2π or $2l$ at equidistant points of x in the interval $(0, 2\pi)$ or $(0, 2l)$ then Fourier coefficients a_0 , a_n and b_n are calculated using the relations given below:

Period	Fourier Coefficients		
2π	$a_0 = \frac{2}{N} \sum y$	$a_n = \frac{2}{N} \sum y \cos nx$	$b_n = \frac{2}{N} \sum y \sin nx$
$2l$	$a_0 = \frac{2}{N} \sum y$	$a_n = \frac{2}{N} \sum y \cos\left(\frac{n\pi x}{l}\right)$	$b_n = \frac{2}{N} \sum y \sin\left(\frac{n\pi x}{l}\right)$

Note:

1. Here **N** denotes the **number of observations**.
2. The amplitude of the first harmonic is given by $\sqrt{a_1^2 + b_1^2}$.
Similarly the amplitudes of the second and third harmonics are given by $\sqrt{a_2^2 + b_2^2}$ and $\sqrt{a_3^2 + b_3^2}$ respectively.

Example:

The following values of 'y' give the displacement of a machine part for the rotation 'x' of a flywheel. Express y as a Fourier series up to first harmonics.

x	0	$\frac{\pi}{3}$	$\frac{2\pi}{3}$	π	$\frac{4\pi}{3}$	$\frac{5\pi}{3}$	2π
y	7.9	7.2	3.6	0.5	0.9	11.7	7.9

Solution:

Here, length of interval is 2π .

Here $N=6$ since $[y]_0^\circ = [y]_{360^\circ}$.

Fourier expansion upto the first harmonic is given by

$$f(x) = \frac{a_0}{2} + (a_1 \cos x + b_1 \sin x)$$

$$a_0 = \frac{2}{N} \sum y$$

$$= \frac{2}{6} \sum y = \frac{1}{3} (26.9) = 8.9667$$

$$a_n = \frac{2}{N} \sum y \cos nx$$

$$\begin{aligned} a_1 &= \frac{2}{6} \sum y \cos x \\ &= \frac{1}{3} \sum y \cos x \end{aligned}$$

$$b_n = \frac{2}{N} \sum y \sin nx$$

$$\begin{aligned} b_1 &= \frac{2}{6} \sum y \sin x \\ &= \frac{1}{3} \sum y \sin x \end{aligned}$$

ENGINEERING MATHEMATICS -II

Practical Harmonic Analysis - Problems



x	y	$\cos x$	$y \cos x$	$\sin x$	$y \sin x$
0	7.9	1	7.9	0	0
$\pi/3$	7.2	0.5	3.6	0.866	6.2352
$2\pi/3$	3.6	-0.5	-1.8	0.866	3.1176
π	0.5	-1	-0.5	0	0
$4\pi/3$	0.9	-0.5	-0.45	-0.866	-0.7794
$5\pi/3$	6.8	0.5	3.4	-0.866	-5.8888
Total	26.9		12.15		2.6846

$$a_0 = \frac{2}{N} \sum y$$
$$= \frac{2}{6} \sum y = \frac{1}{3} (26.9) = 8.9667$$

$$a_n = \frac{2}{N} \sum y \cos nx$$

$$a_1 = \frac{2}{6} \sum y \cos x$$
$$= \frac{1}{3} \sum y \cos x$$
$$= \frac{1}{3} [12.15] = 4.05$$

$$b_n = \frac{2}{N} \sum y \sin nx$$

$$b_1 = \frac{2}{6} \sum y \sin x$$
$$= \frac{1}{3} \sum y \sin x$$
$$= \frac{1}{3} [2.6846] = 0.8949$$

Therefore Fourier series up to first harmonic is given by

$$f(x) = \frac{a_0}{2} + a_1 \cos x + b_1 \sin x$$

$$y = 4.48335 + 4.05 \cos x + 0.8949 \sin x$$



THANK YOU

Dr. G K Jagatheswari

Department of Science & Humanities

jagatheswarigk@pes.edu